Calling Python from Go



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Extending and Embedding the Python Interpreter

Extending

Create a shared library in a specific format, use it from Python.

- C
- Cython
- Rust (PyO3)
- ...

Embedding

Python is a shared library with well documented API.

The Python executable is a thin(?) wrapper around this shared library.

Cgo anyone?

The Other Option - RPC

Run Python as a external process and communicate with it.

Pick your RPC: stdin/stdout, HTTP, gRPC, ...

Pay the serialization and transport price.

Pick Your Poison

Embedding	RPC
Fast: function call, can avoid some serialization	Slow: Serialize and transport.
Cgo	Pure Go
Same process, shared memory	Isolated process

Types

Python & Go have different types.

- Python's set
- Go's int8, int16, int32, ...

Serialization format brings its own types.

Dependencies

You need Python runtime to run the code.

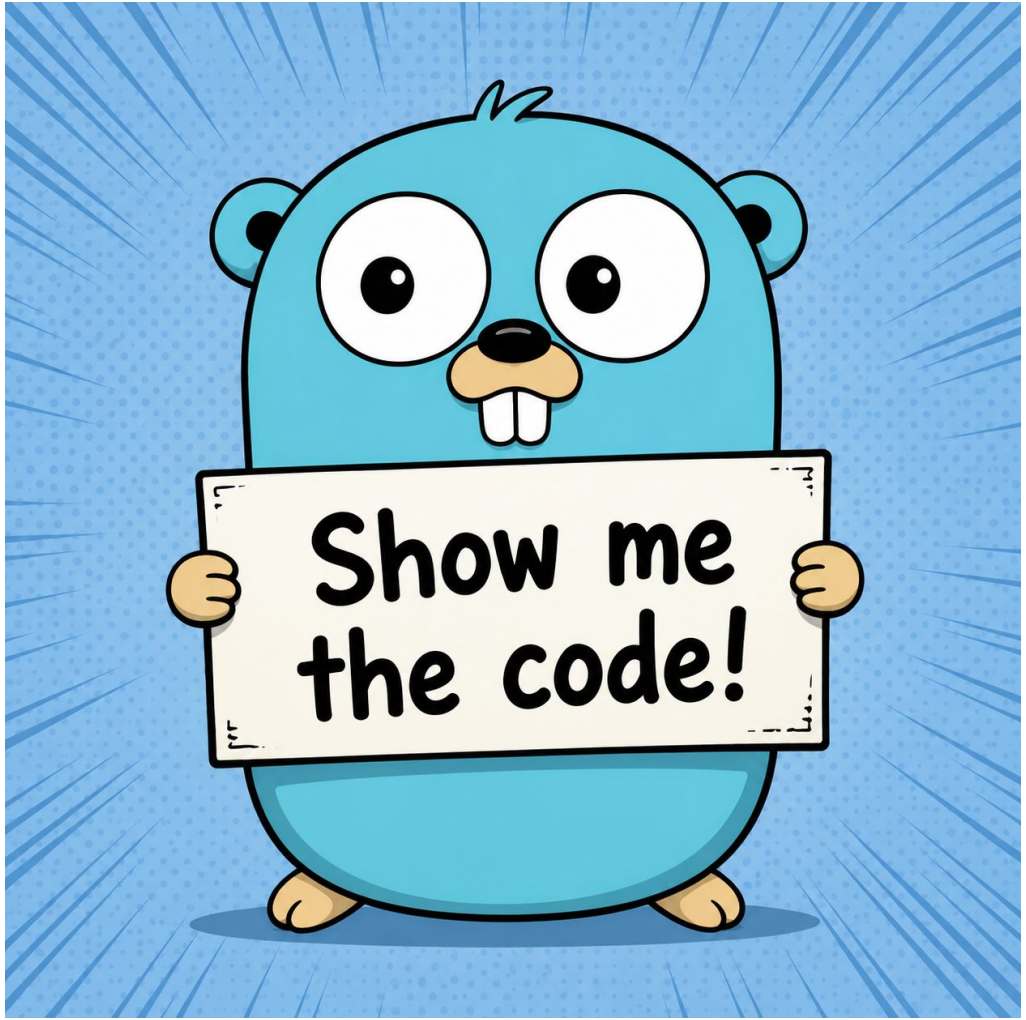
If you use external Python packages, you need to install them as well.

Example: Outlier Detection

Python implements outlier detection.

Return indices of values that are more than 3 std than the mean.

We'll use the numpy package.

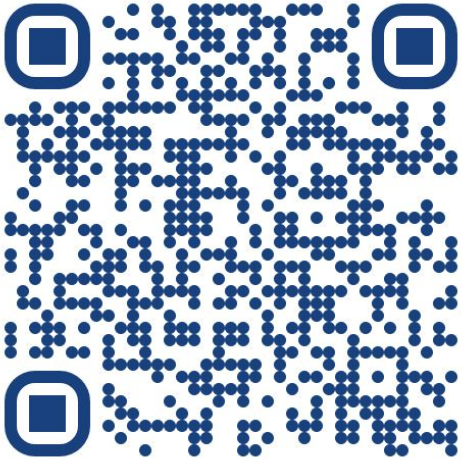

Show me
the code!

Conclusion

Pick RPC if you like to sleep at night.

Pick embedding if you need speed, love long stack traces and like to learn about `runtime.AddCleanup` at 4am.

Thank You



github.com/tebeka/talks/tree/main/go-to-py